Chapter One

Introduction to Internet
and the World Wide Web

The Internet

- Network of networks that use the Internet protocol suite to link billions of devices worldwide.
  - *The Internet is a global system of interconnected computer networks*
- Consists of millions of private, public, academic, business, government networks.
- Networks linked together by electronic, wireless, & optical networking technologies
- Carries information resources and services, e.g. WWW

Brief history of Internet

- Began as a US Department of Defense network called ARPANET (1960s-70s)
  - **ARPANET** Main idea was to
    - support packet switching – allowing multiple users to use same communication path and
    - Use TCP (transmission control protocol) to ensure message is properly pass from sender to receiver
- Initial services: electronic mail, file transfer
- Opened to commercial interests and most universities in late 80s
- **WWW** created in 1989-91 by Tim Berners-Lee
- Early web browsers released: **Mosaic** 1992, **Netscape** 1994, **Internet Explorer** 1995
- **Amazon**.com opens in 1995; **Google** January 1996

The Web

- In 1989,Tim Berners-Lee of CERN (the European Organization for Nuclear Research) began to develop a technology for sharing information via hyperlinked text documents.
- He also wrote *communication protocols (Hypertext Transfer Protocol(HTTP))* to form the backbone of his new information system, which he called the World Wide Web.
- The **Web (World Wide Web)** consists of information organized into Web pages containing text and graphic images.
- It contains hypertext links, or highlighted keywords and images that lead to related information.
- A collection of linked Web pages that has a common theme or focus is called a **Web site**.

Web fundamentals

- The WWW comprises Web Servers and Web Browsers
  - **Web Server**: software that listens for Web page requests and serves up the requested pages
    - Apache - http://www.apache.org
    - Microsoft Internet Information Server (IIS) - http://www.iis.net/
    - Express - https://expressjs.com
  - **Web browser**: gets and renders documents from servers

Web fundamentals

- Hyperlinks: used to reference other web pages, e-mail addresses, files and more…
- URIs (**Uniform Resource Identifiers**)identify resources on the Internet
- **URIs** that start with **http://**are called **URL**s (**Uniform Resource Locators**)
- **HTTP –**hypertext transport protocol
  - application-level protocol for distributed, collaborative, hypermedia information systems
  - request/response protocol

HTTP

- Defines a set of commands understood by a Web server and sent from a browser
- Some HTTP commands (your browser sends these internally)
  - GET resource -- requests data from a specified resource
  - POST resource -- submits data to be processed to a specified resource
  - PUT resource -- uploads a representation of the specified URL
  - DELETE resource -- deletes the specified resource
- Reading: HTTP status codes

Internet Media Types (MIME)

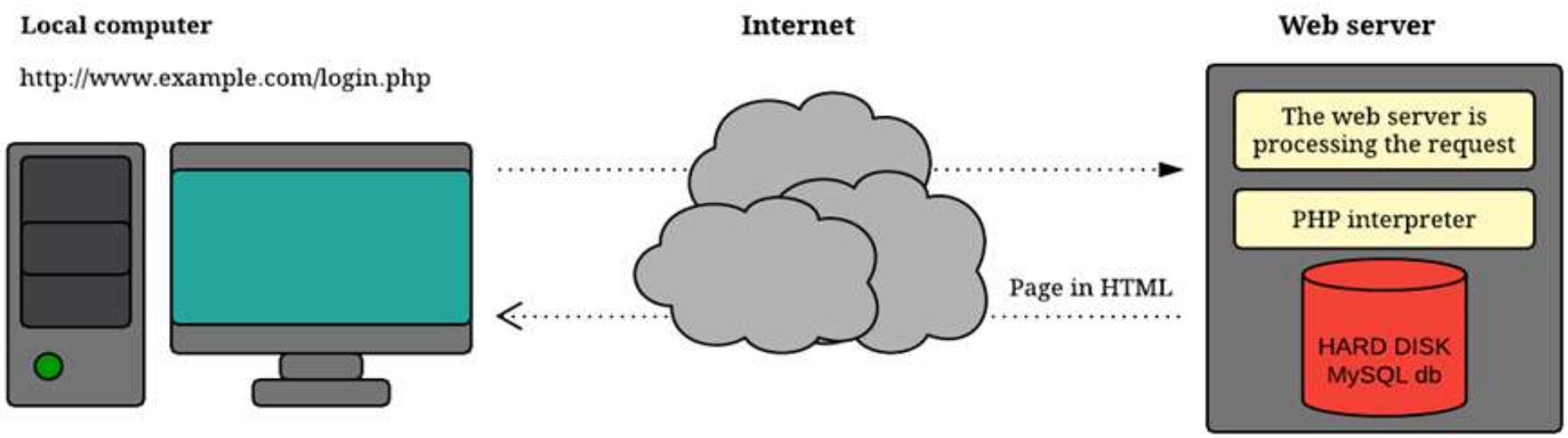
- **Sometimes when including other resources in a Web page (stylesheet, image, multimedia object), we specify their type of data**

MIME Type	File Extension
text/html	.html
text/plain	.txt
image/gif	.gif
image/jpeg	.jpg
videeo/quicktime	.mov
application/octec-stream	.exe

Web Client and server

- **Client = User agent**. It is controlled by a user to operate the Web application.
- A **Web browser** is the software that you run on your computer to make it work as a web client.

Web server: A Web server is a software that supports various Web protocols like HTTP, and HTTPS, etc., to process client requests.

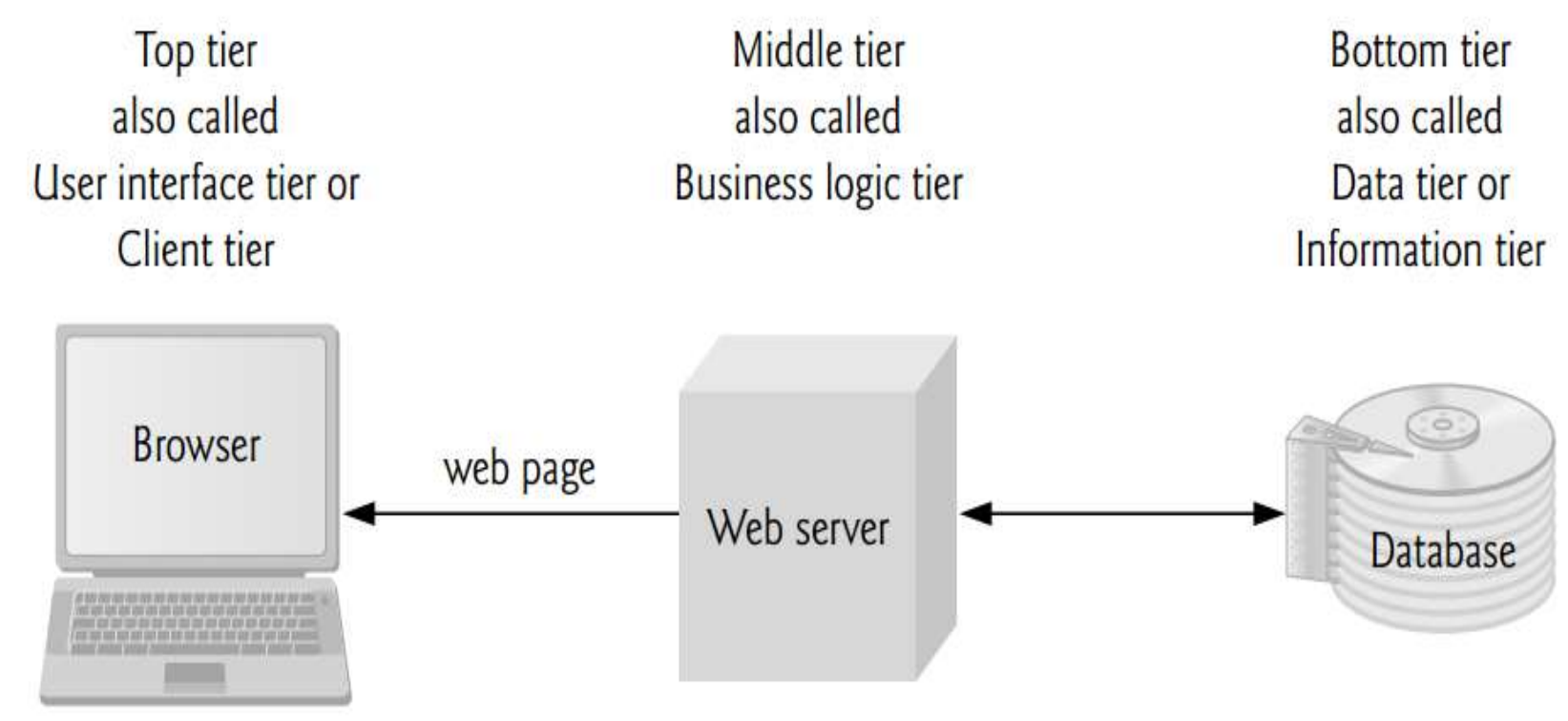


Web Server

- Web servers are computer systems that deliver content or services to end users over the internet.
- A Web server is also known as an Internet server.
- It consists of a physical server, server operating system (OS) and software used to facilitate HTTP communication.
- Example: Apache, IIS, Tomcat

Multi tier application

- **Web-based applications are often multi-tier applications(sometimes referred to as n-tier applications) that divide functionality into separate tiers(i.e., logical groupings of functionality)**

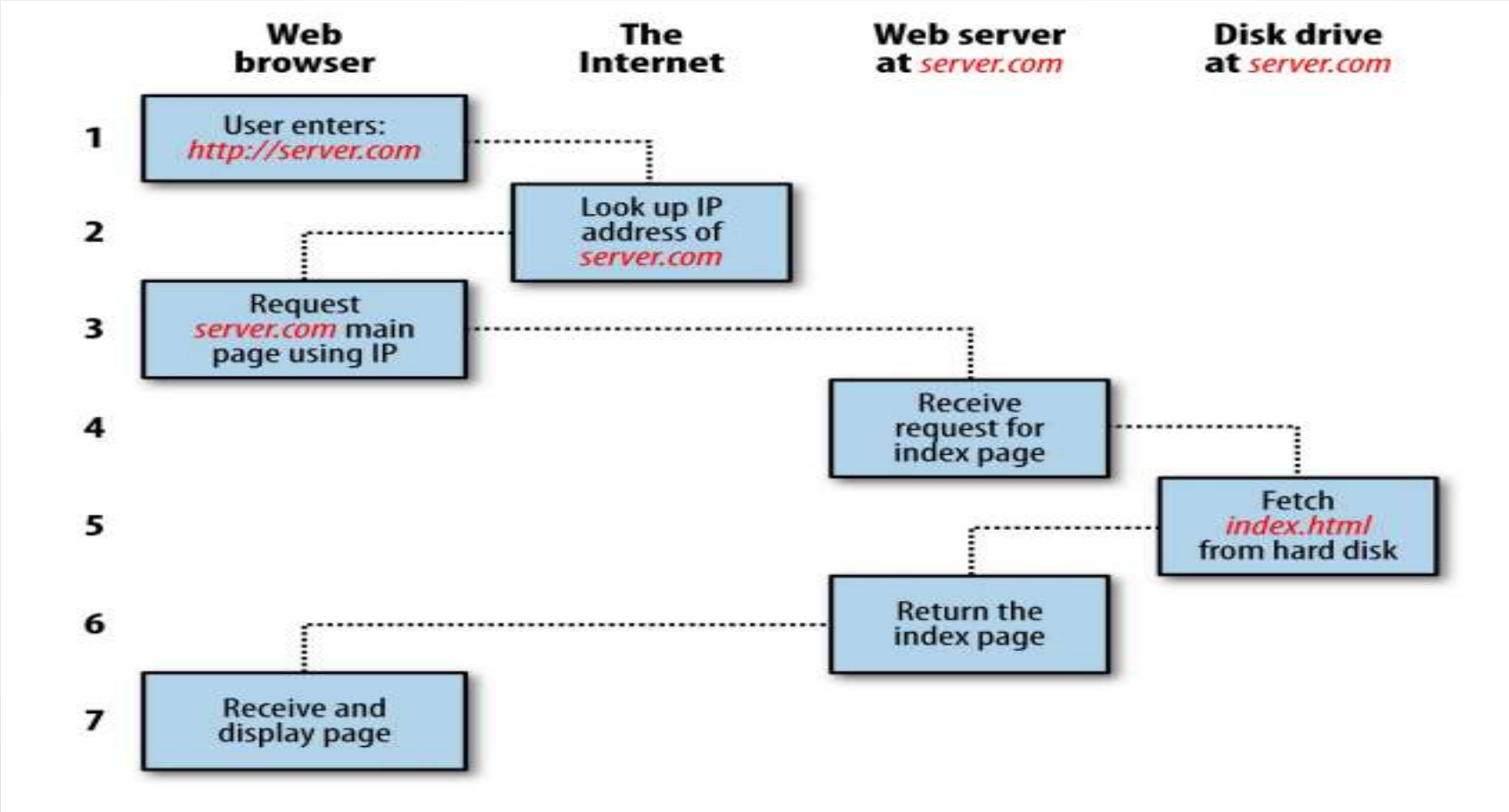


Multi tier…

- The **bottom** tier(also called the data tier or the information tier) maintains the application's data. This tier typically stores data in a relational database management system (RDBMS).
- The **middle** tier implements business logic, controller logic and presentation logic to control interactions between the application's clients and its data.
- The middle-tier controller logic processes client requests (such as requests to view a product catalogue) and retrieves data from the database.
- The middle-tier presentation logic then processes data from the information tier and presents the content to the client.
- Web applications typically present data to clients as HTML documents.
- Business logic in the middle tier enforces business rules and ensures that data is reliable before the application updates a database or presents data to users.
- Business rules dictate how clients access data and how applications process data.
- The **top** tier, or client tier, is the application's user interface, which gathers input and displays output.

Request response procedure

- The basic client/server request/response sequence



Request response…

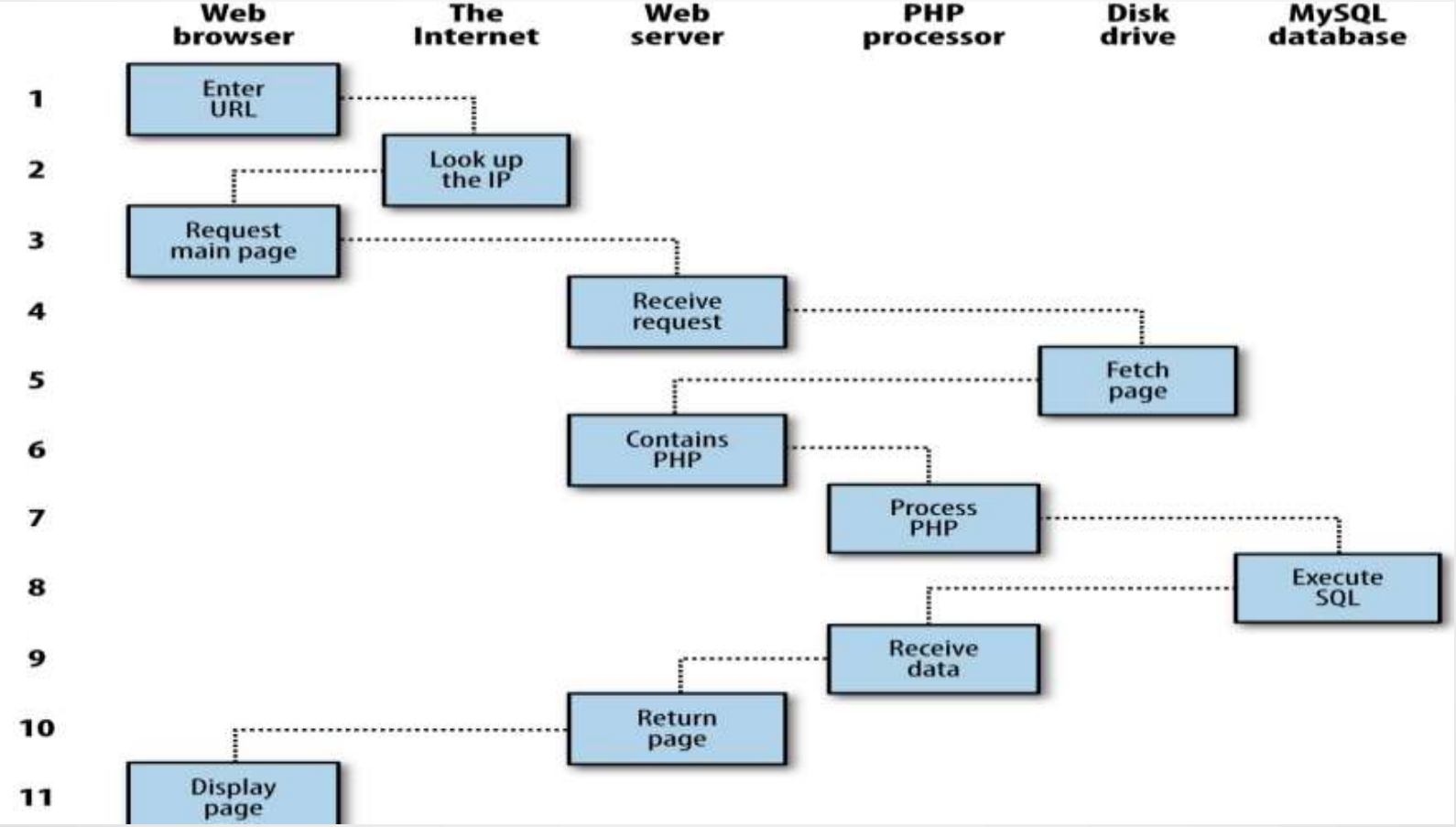
- Each step in the request and response sequence is as follows:

1. You enter http://server.cominto your browser's address bar.
2. Your browser looks up the IP address for server.com.
3. Your browser issues a request for the home page at server.com.
4. The request crosses the Internet and arrives at the server.com web server.
5. The web server, having received the request, looks for the web page on its disk.
6. The web page is retrieved by the server and returned to the browser.
7. Your browser displays the web page.

- For an average web page, this process takes place once for each object within the page: a graphic, an embedded video or Flash file, and even a CSS template.

Request response…

- A dynamic client/server request/response sequence



Request response…

- You enter http://server.com into your browser's address bar.
- Your browser looks up the IP address for server.com.
- Your browser issues a request to that address for the web server's home page.
- The request crosses the Internet and arrives at the server.com web server.
- The web server, having received the request, fetches the home page from its hard disk.
- With the home page now in memory, the web server notices that it is a file incorporating
- PHP scripting and passes the page to the PHP interpreter.
- The PHP interpreter executes the PHP code.
- Some of the PHP contains SQL statements, which the PHP interpreter now passes to the MySQL database engine.
- The MySQL database returns the results of the statements to the PHP interpreter.
- The PHP interpreter returns the results of the executed PHP code, along with the results from the MySQL database, to the web server.

The HTTP Request Circle

- A typical HTTP request / response circle: [*www.w3school.com*]
- The browser requests an HTML page. The server returns an HTML file.
- The browser requests a style sheet. The server returns a CSS file.
- The browser requests an JPG image. The server returns a JPG file.
- The browser requests JavaScript code. The server returns a JS file
- The browser requests data. The server returns data (in XML or JSON).
-

Standard Web Technologies

- World Wide Web Consortium (W3C) (https://www.w3.org/)
- international consortium that develops and publishes open standards
- aims to ensure long-term growth of web
- recommends a list of standard web technologies:
  - HTTP
  - HTML5
  - CSS3
  - JavaScript
  - DOM
  - PHP
  - MySQL

HTML

- The public files on the web servers are ordinary text files, much like the files used by word-processing software.
- To allow Web browser software to read them, the text must be formatted according to a generally accepted standard.
- The standard used on the web is Hypertext markup language (HTML).
- HTML uses codes, or tags, to tell the Web browser software how to display the text contained in the document.

HTML…

- HTML stands for **HyperText** Markup Language:
- A ***markup*** language is a computer language that defines the structure and presentation of raw text.
- In **HTML**, the computer can interpret *raw text* that is wrapped in **HTML** elements.
- ***HyperText*** is text displayed on a computer or device that provides access to other text through links, also known as ***hyperlinks***.

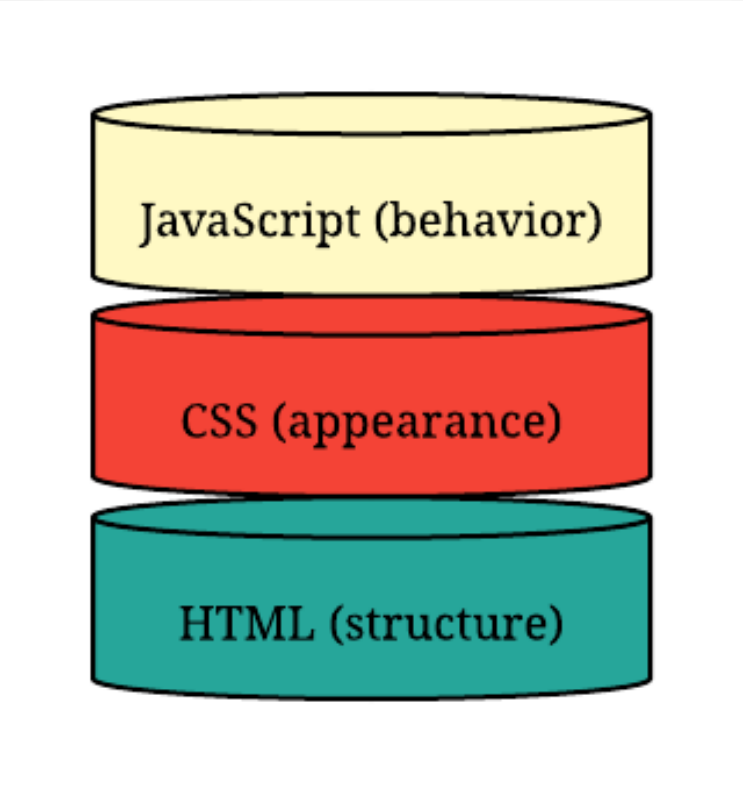
CSS

- Cascading Style Sheets
- Simple way to add style to your html and web documents
- add colour, whitespace, font, styling
- makes it possible
- to separate the content from the presentation
- share presentation rules across multiple pages
- make accessibility easier to achieve
- make responsive design easier

JavaScript

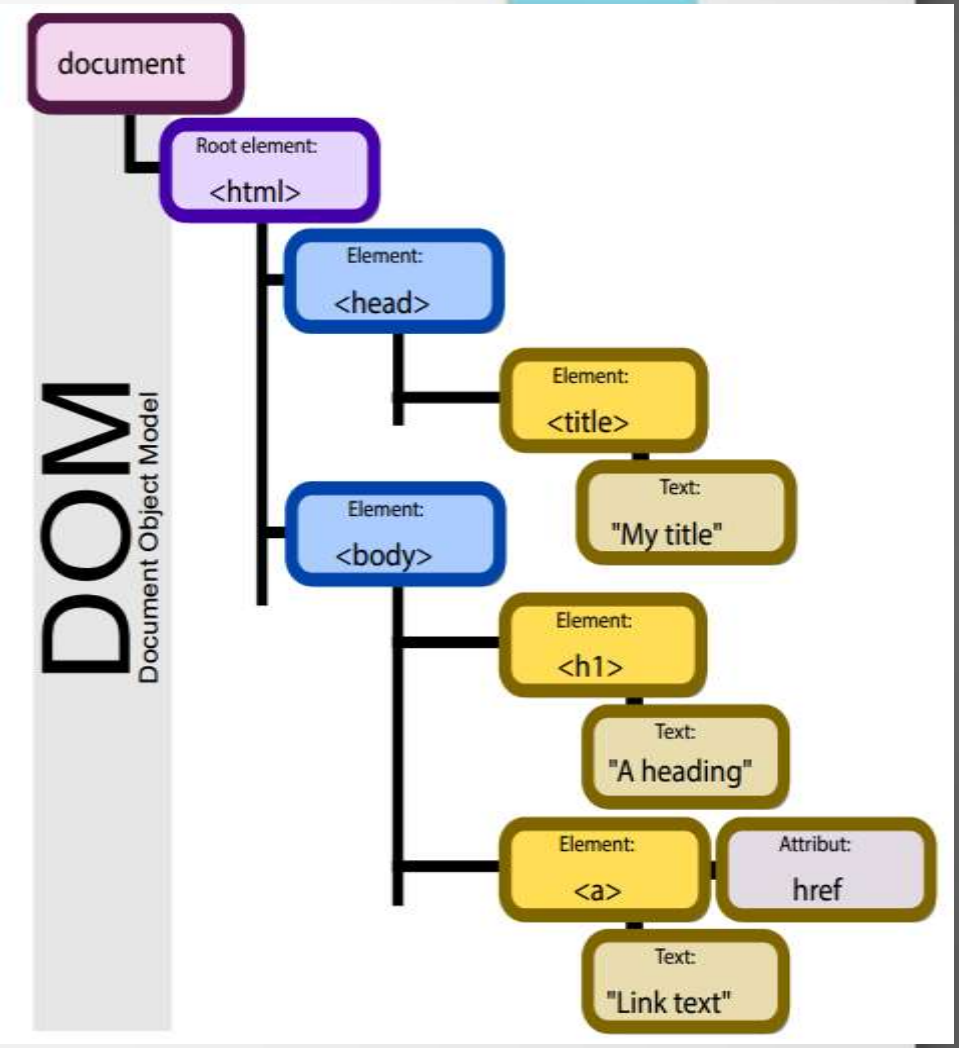
- Client-side scripting language (mainly)
- shares some syntax with Java, but is a distinct language
- Imperative, interpreted, dynamically-typed, object oriented
- JavaScript can do
  - JavaScript Can Change HTML Content
  - JavaScript Can Change HTML Attribute Values
  - JavaScript Can Change HTML Styles (CSS)
  - JavaScript Can Hide HTML Elements
  - JavaScript Can Show HTML Elements

How the client side works



DOM

- platform-neutral and language-neutral interface
- lets programs/scripts dynamically access:
- content
- structure
- Style
- Treats XML, HTML, XTML document like a tree



Server side scripting: PHP

- PHP: Hypertext Preprocessor
- used mainly for web development
- can be embedded in HTML
- typically interpreted
- used to dynamically generate content
- unlike JavaScript, is server-side

```
<body>
    <?php echo '<p>Hello World!</p>'; ?>
</body>
```

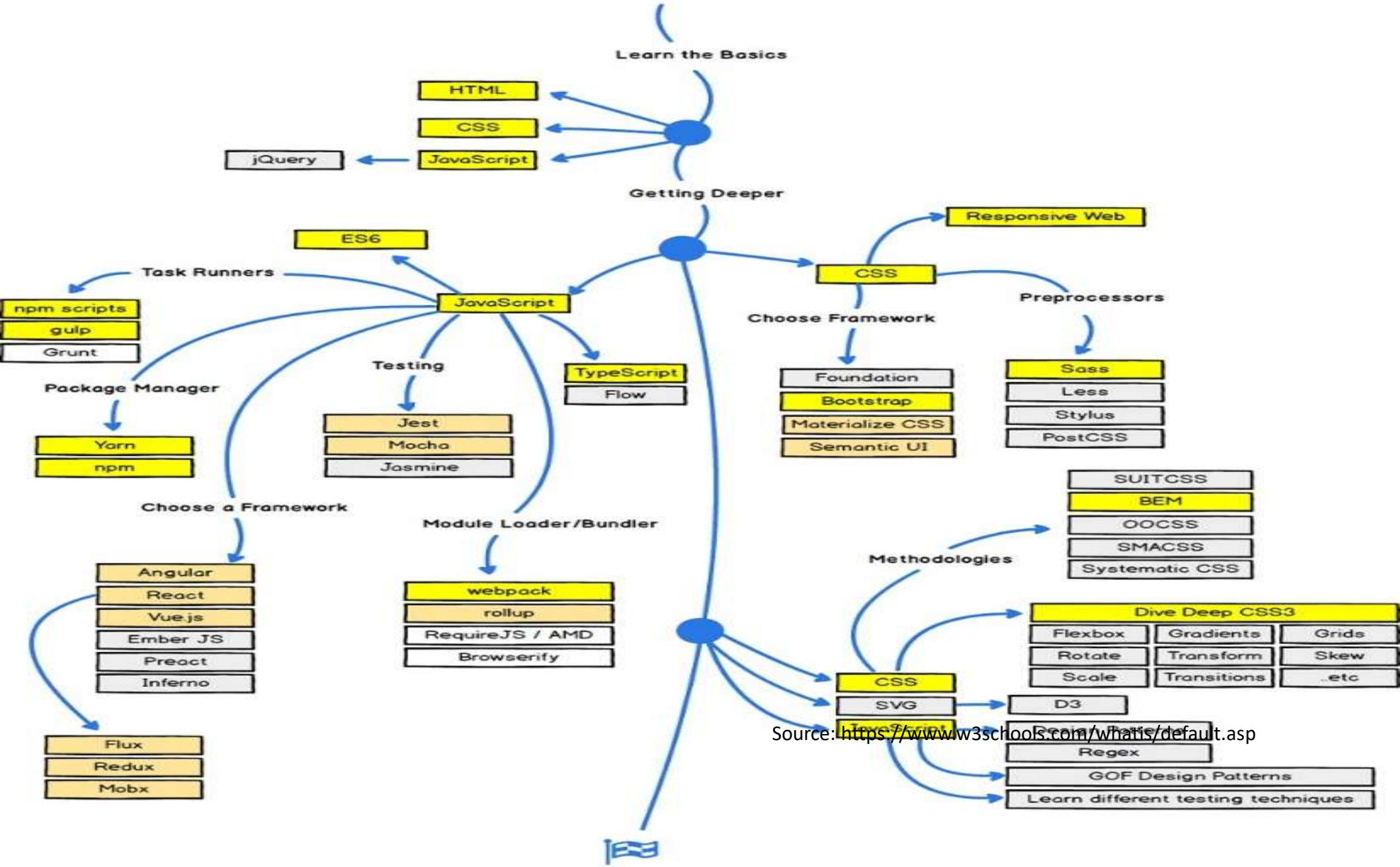
Web Development Principles

- It is important to create websites that are:
- attractive
- functional
- user-friendly
- Start with
  - what is the website?
  - what is the audience?
- Design navigation based on audience and purpose
- Provide multiple links from different contexts to vital parts of site

Web Development …

- Keep it Simple Stupid (KISS)
- avoid lots of video and audio
- choose familiar and appropriate
  - fonts
  - colours
- use contrast and whitespace
- ensure fast load
- use a fluid layout for different screen sizes
- use open standards
- test your site on many browsers

Front- end road map



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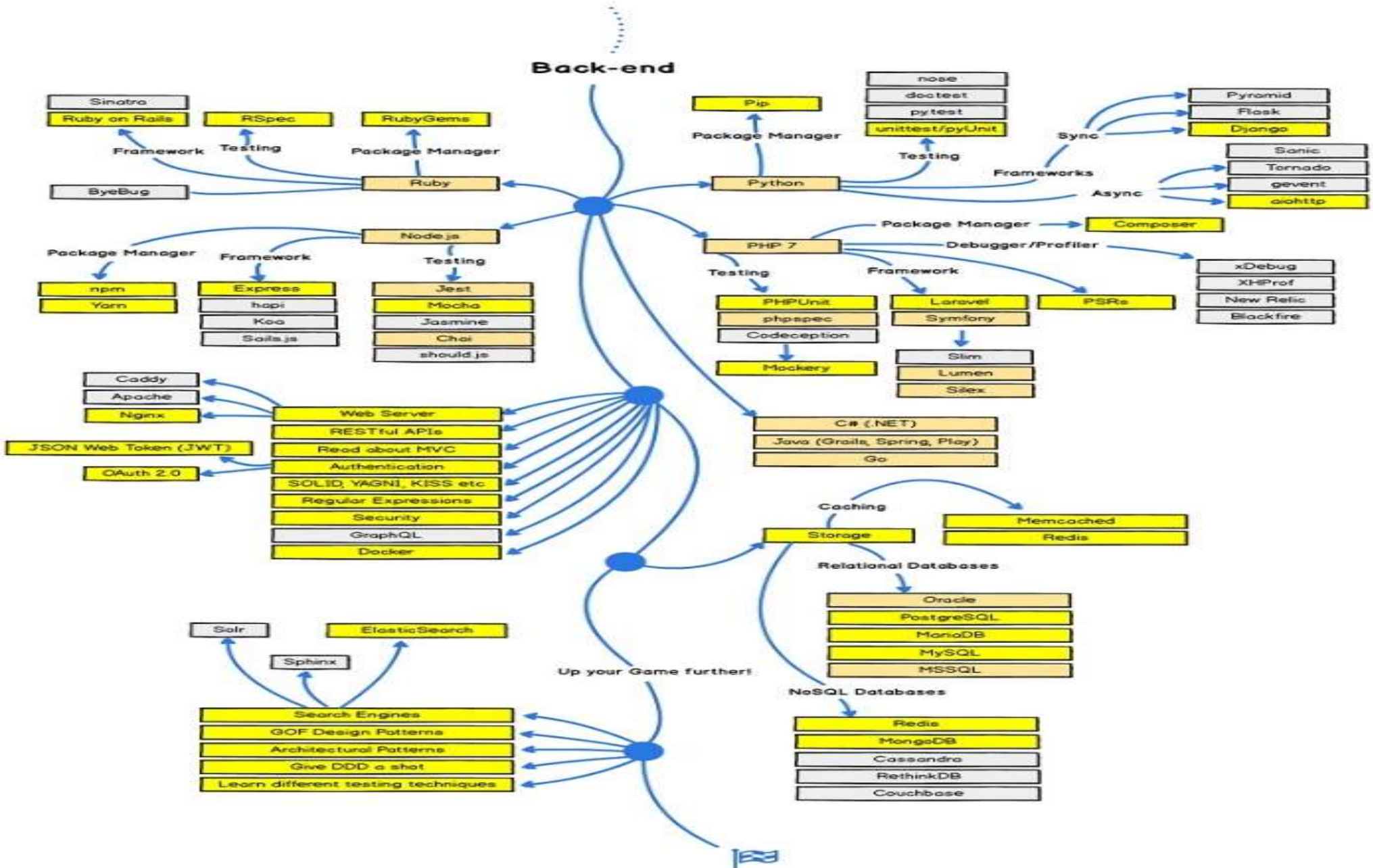
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